Digital Signal Processing

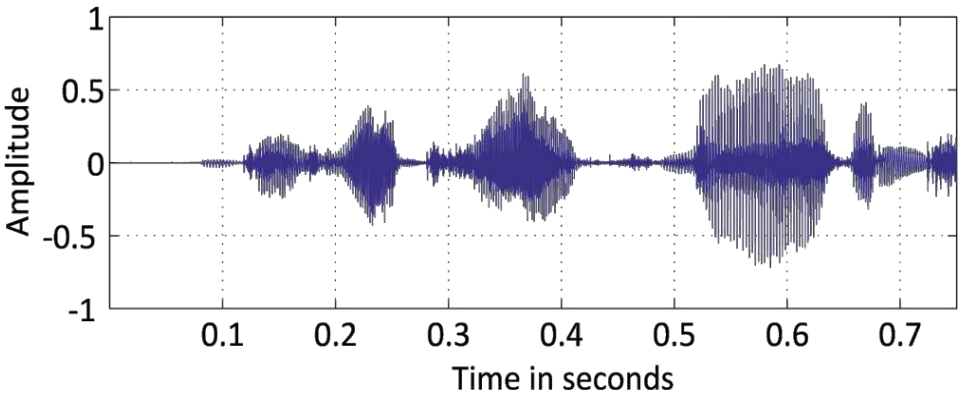
Lecture # 01 Topic: Signals

Text Books

- Signals and Systems
  - Alan V. Oppenheim, Alan S. Willsky, S. Hamid Nawab, Prentice Hall Ltd. Second edition
- Signals and Systems
  - Simon Haykin, Barry Van Veen, John Wiley & Sons Inc.
- Signals and Systems
  - M. J. Roberts, McGraw-Hill Edition
- Analysis of Linear Systems
  - David K. Cheng, Narosa Publishing House

What is a signal?

- A signal is a varying quantity whose value can be measured and which conveys information
- A signal is a physical quantity that varies with time, space, or any other independent variable
- Mathematically: A signal is a function of one or more independent variables, $v(t) = 5t$, $v(n) = 20n^2$,
- Example: temperature over time, $T(t)$, brightness (luminance) of an image $I(x, t)$, the pressure of a sound wave over time $p(x, y, z)$ or $p(x, y, z, t)$
- Example: temperature
  - It can vary over time
  - We can measure it using a thermometer
  - It conveys information: knowing the temperature outside will inform our decision as to which clothes to wear
- In the digital signal processing system, the signal is represented as a sequence of numbers either on a computer or in digital hardware
- Example:
  - we could store the temperature at various times of the day as a sequence of numbers in an array on a computer: each reading might be a temperature reading in Celsius



Examples of Signals

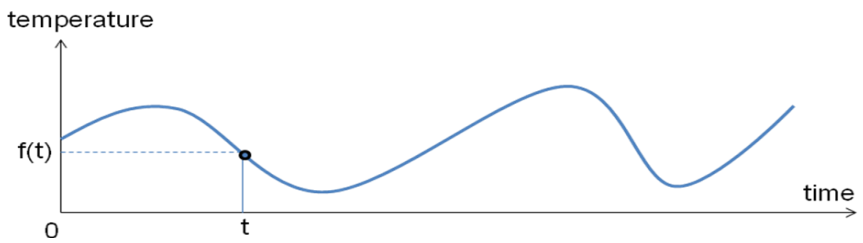
- Signals are functions of one or more variables (independent variables) that carry/convey information.
- For example:
  - Electrical signals - voltages and currents in a circuit, power
  - Thermal signals – temperature
  - Light signals – light intensity,
  - Mechanical signals – force, torque, pressure
  - Acoustic signals ---audio or speech signals (analog or digital) such as human voice, a dog's bark, a bird's song,
  - Video signals ---intensity variations in an image (e.g., a CAT scan, MRI data)
  - Biological signals ---sequence of bases in a gene
  - Human Body Signal – ECG, EMG, EOG; Brain signal - EEG signal
  - Stock market data

The domain of a Signal

- The domain is a very widely used term in DSP.
  - For instance, a signal that uses time as the independent variable (i.e., the parameter on the horizontal axis), is said to be in the time domain
  - Another common signal in DSP uses frequency as the independent variable, resulting in the term, frequency domain.
  - Likewise, signals that use distance as the independent parameter are said to be in the spatial domain (distance is a measure of space).
  - Sometimes space (x, y) is the spatial domain, and (x, y, t) is the spatiotemporal domain.

Domain and range of a signal

- Temperature is a function of single real-valued variable, T=f(t)
- We say that the ***domain*** of the signal is one-dimensional



- Some signals are functions of more than one variable

- *For example*: a black and white photograph – can be regarded as a signal
  - Brightness u of a point on the photograph is a function of two variables, the x and y coordinates of the point on the photograph
  - The domain of the signal is two-dimensional (fig. 2)
  - The range of the signal is one-dimensional

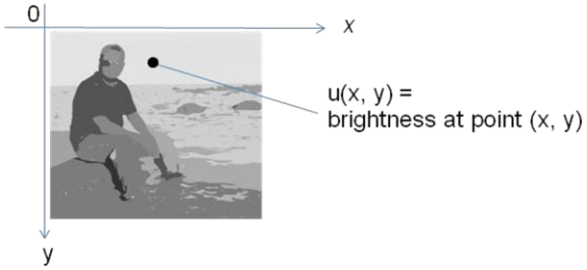


Fig. 2

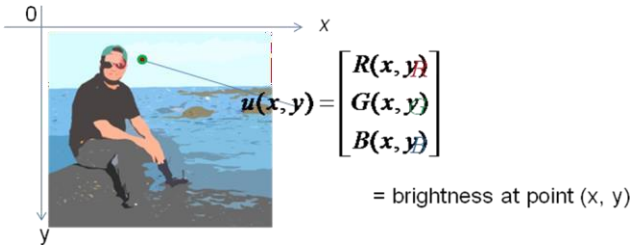


Fig. 3

Signals	**Domain**	**Range**
Black and white photograph	Domain of the signal is two-dimensional	Range of the signal is one-dimensional
Color photograph	***domain*** is two-dimensional	The color of a point on the photograph can be expressed as three real numbers, especially given the amount of red (R), green (G), and blue (B) that go to make up the color. In this case, we could say that the ***range*** of the signal is three-dimensional
Color Movie / Cinema	Domain is 3D (x, y, and time)	***range*** of the signal is three-dimensional
Voltage, Current, power	1D (t)	1D
ECG, EMG, EEG, EOG	1D (t)	1D
Audio, speech	1D (t)	1D
Black & White movie	3D (x, y, t)	1D
Temperature	1D (t)	1D
Velocity	1D (t)	1D
Pressure	p(x, y, z, t)	1D
MRI	P(x, y, z)	1D
CT Scan	P(x, y, z)	1D

The dependent variable of a signal

- The dependent variable of a signal
  - Examples: Specific: voltage, light intensity, sound pressure, or an infinite number of other parameters
  - Generic label: amplitude
- Another name for the Dependent variable
  - vertical axis
  - y-axis,
  - range, and
  - ordinate

Independent variable of a signal

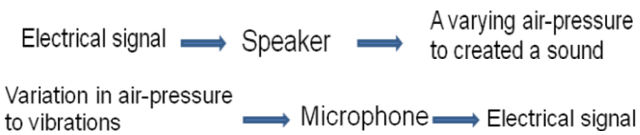
- The independent variable of a signal
  - Most common parameter: Time
  - Distance
  - x
  - Number
  - Generic name: sample number
- Another name for the independent variable
  - horizontal axis
  - x-axis,
  - the domain, and
  - the abscissa

Dependent and independent variables of a signal

- Dependent variables
  - Can be continuous
    - Speech, current, voltage, temperature, audio
  - Can be discrete
    - DNA base sequence
    - Digital image pixels
  - Can be 1-D (speech signal), 2-D (image signal), ••• N-D
- Independent variable
  - Single (1-D) independent variable which we call" time”.
  - Continuous-Time (CT) signals: x(t), t—continuous values
  - Discrete-Time (DT) signals: x[n], n—integer values only

Converting a signal from one form to another

- A device that converts a signal from one form to another is called a transducer
- Example: Loudspeaker, microphone



- Variation of air pressure is a real-valued function of time

What is Noise? How can noise be represented?

- Noise is a random signal, is like a signal in that it is a time-varying physical phenomenon, but unlike a signal, it usually does not carry useful information and is almost considered undesirable

- Noise Representation: Noise can be represented as a function of time like a signal

  The function of time t: $n(t) = a(t)[\cos(2\pi ft + \Theta(t))]$;

$a(t)$ and $\Theta(t)$ are time-variant amplitude and phase of the noise; $\Theta(t) \in [-\pi, \pi]$

What does "Digital Signal Processing" mean?

The term "Signal Processing" in "Digital Signal Processing" is the passing of the signal through a system

Examples:

(i) Modification of the signal (filtering, interpolation, noise reduction, equalization, …)

(ii) Prediction, transformation to another domain (e.g., Fourier transform)

(iii) Numerical integration and differentiation

(iv) Determination of mean value, correlation, probability density function, …

Properties of the system (e.g., linear/nonlinear) determine the properties of the whole processing operation.

The definition of a system also includes:

(i) Software realizations of operations on a signal, which are carried out on a digital computer (software implementation of the system),

(ii) digital hardware realizations (logic circuits) configured such that they can perform the processing operation, or

(iii) most general definition: a combination of both.

What do you mean by signal processing task? Give some examples.

- Modify the number of bass and treble in an audio signal
- Analyze the image to determine what objects are present in it
- Compute seasonally adjusted temperature values
- Make a photograph sharper and increase its contrast
- Measure the pitch of a musical instrument
- Compression of data for transmitting and store

Analog vs. digital signal/data

- Data can be analog or digital. The term analog data refers to information that is continuous; digital data refers to information that has discrete states. Analog data take on continuous values. Digital data take on discrete values.

- Analog and Digital data
  - Data can be analog or digital.
  - Analog data are continuous and take continuous values.
  - Digital data have discrete states and take discrete values.

- Analog and Digital Signal
  - Signals can be analog or digital.
  - Analog signals can have an infinite number of values in a range.
  - Digital signals can have only a limited number of values

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